QP CODE **Reg.No :**

Name :

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION

THIRD SEMESTER

MG3DSCPHY200 – Principles of Mechanics

(2024 ADMISSION ONWARDS)

Duration: 1.5 hrs Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions
Answer any **10** questions
Each question carries **2** Mark

1	Explain the Law of Inertia.	[U][1]
2	Write down 2 equations of motions for the situation in which force is constant.	[A][1]
3	A force F(t) = 2t N acts on a 1 Kg mass. If the object starts from rest, find its velocity at t = 3 seconds.	[A][1]
4	What is the equation representing Newton's law of gravitation? Explain the terms.	[U][2]
5	What do you mean by the gravitational field?	[U][2]
6	Plot the variation of potential and Kinetic energy with displacement of a harmonic oscillator	[A][3]
7	Write the difference between the motion of critically damped and underdamped oscillation.	[U][4]
8	What are the factors that affect the amplitude of a forced harmonic oscillator?	[U][5]
9	What is the difference between linear momentum and angular momentum?	[A][6]
10	A skater is spinning on the ice.They are initially spinning slowly with their arms outstretched. They then pull their arms in, causing them to spin much faster. Is the skater's kinetic energy conserved in this process? Explain.	[A][6]
11	Express the angular momentum of a system of particles in terms of angular momentum of centre of mass. Explain the terms.	[An][6]
12	In a symmetric rigid body rotating about a fixed axis, what happens to the perpendicular component of angular momentum? Explain.	[An][6]
13	State and explain Kepler's second law.	[U][7]
14	What are the features of central force? Give two examples of central forces.	[An][7]

(2x10=20)

Part B

Short Essay Type Questions
Answer any **6** questions
Each question carries **5** marks

15	How will you determine the motion of a particle if the force F is a function of the speed of the particle?	[U][1]
16	What is meant by Gravitational Potential. Explain the Physical Significance.	[U][2]
17	Particle executing simple harmonic motion has a maximum velocity 50 m/s and maximum acceleration 80 m/S^2 . Calculate its amplitude and frequency.	[A][3]
18	Solve the differential equation of a damped harmonic oscillator. Discuss in detail about the underdamped oscillator.	[U][4]
19	Write an equation of motion and its solution for a forced damped oscillator . Explain the term resonance.	[U][5]
20	A 1 kg stone at the end of a 2 m long string makes 5 revolutions per second. Calculate the force on the stone as measured in an inertial frame and in a frame which is rotating with the string.	[A][6]
21	Calculate the effective weight of an astronaut ordinarily weighing 60 kg,when his rocket moves vertically (a) upward (b) downward with 6 g acceleration	[An][6]
22	Explain the concept of effective potential. Draw the plot of Effective potential versus radial distance from centre of force for the Kepler potential.	[U][7]

(5x6=30)